CMPE 297-01 Computer Vision

Introduction to Computer Vision

Fall 2026, DMH 348

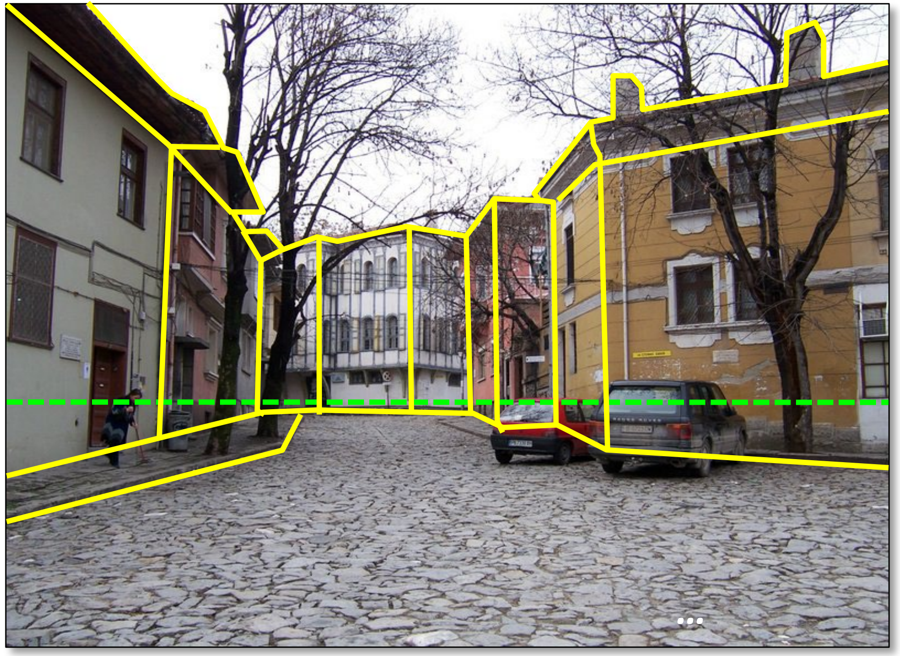
Lecture Outline

- Goal of computer vision
- Origins of computer vision
- Topics covered in class

Goal of Computer Vision

Goal: Image understanding

Goal: Image understanding



Geometric understanding

Goal: Image understanding

Semantic understanding

Goal: Image understanding

Goal: Image understanding

Affordances

Goal: Image understanding

Actionable information

Goal: Image understanding

This image depicts a quiet, cobblestone street in what appears to be an older European town or city. The street is lined with historic buildings that have a slightly weathered appearance, suggesting that they have been standing for many years.

On the left side of the image, there is a person sweeping the street near a light green building. The buildings on this side have old wooden windows and are painted in muted colors, such as light green and a faded pink. The walls show signs of age, with some cracks and peeling paint.

The right side of the street features buildings with a different style and color scheme. One building, in particular, is painted a pale yellow, with visible signs of wear and tear on its façade. The corner of this building is curved, giving it a distinctive look. There are two parked cars in front of this building, one red and the other gray.

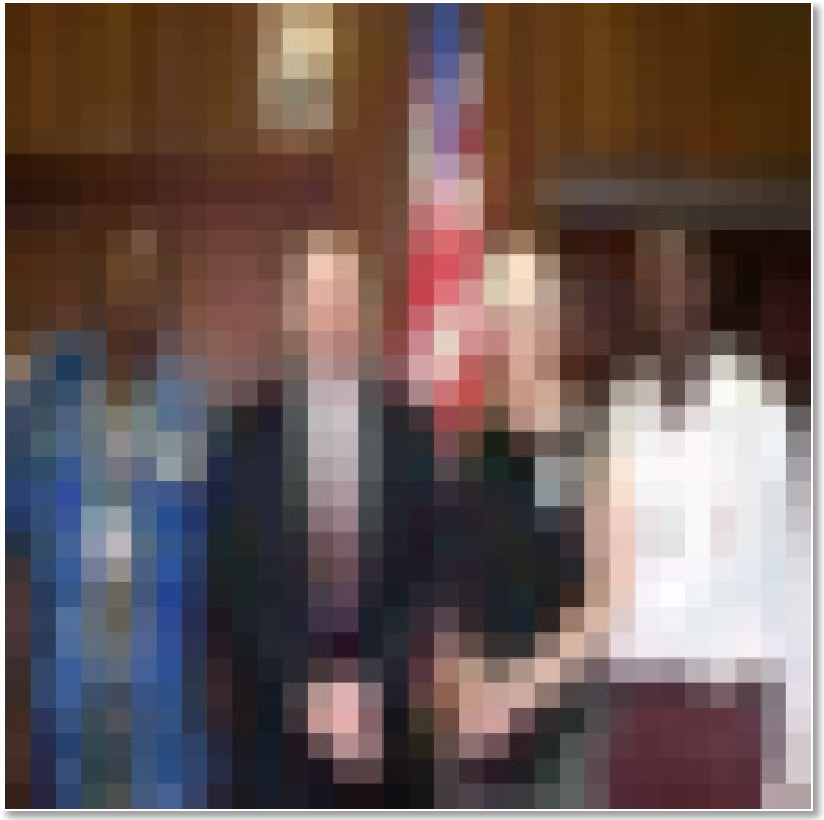
In the background, directly ahead, there is a larger, more ornate building painted white with several rows of tall, narrow windows. This building has a historical and architectural significance, with its unique design and intricate window details. The street gently curves around this building.

The trees lining the street are bare, indicating that the photo was taken during late autumn or winter. The overall mood of the image is quiet and serene, capturing a moment in time in an older part of town.

Now generate an image using the above text as a prompt

Image generation

Humans are quite good at image understanding



Source: “80 million tiny images” by Torralba et al.

Humans are quite good at image understanding

(most of the time)

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Figure from Marr (1982), attributed to R. C. James

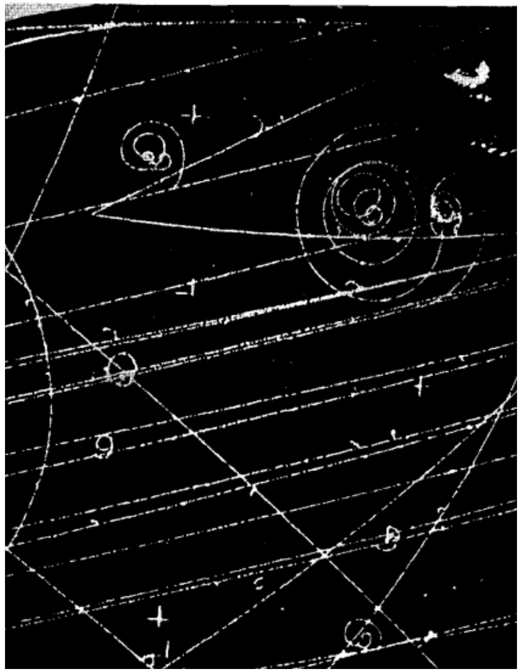
Humans are quite good at image understanding

(most of the time)

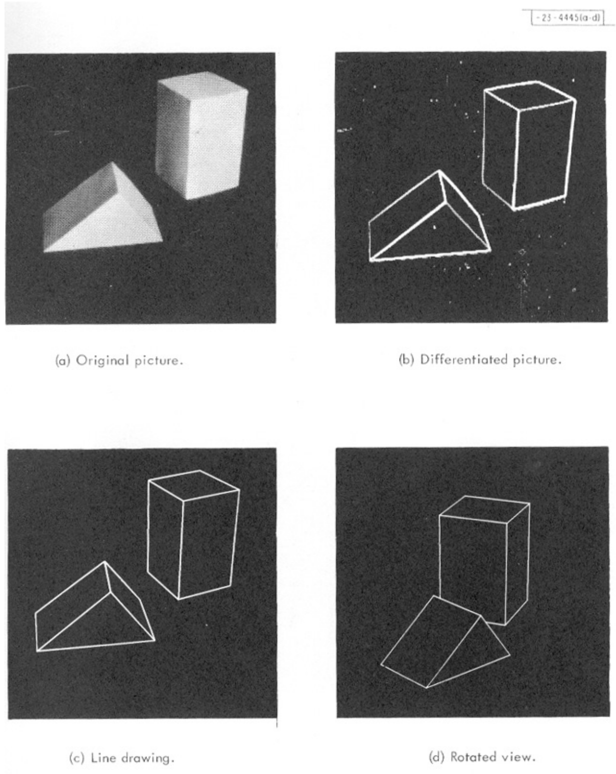
Remember: Images are fundamentally ambiguous!

Origins of Computer Vision

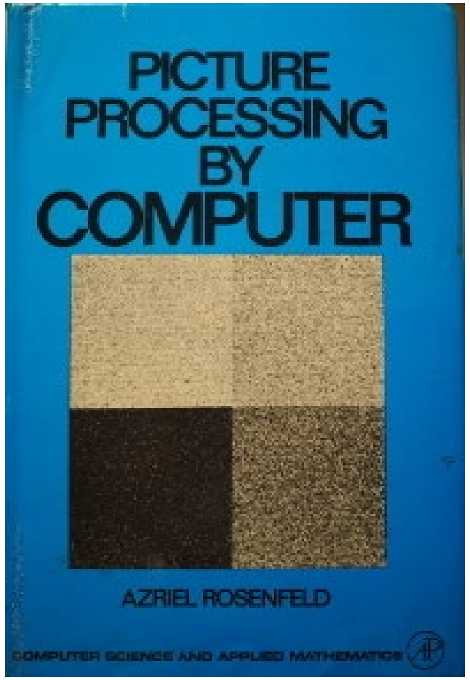
Origins of computer vision



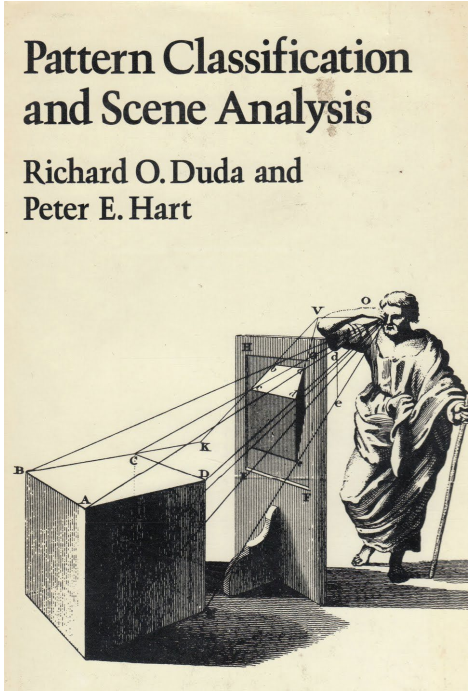
Hough, 1959



Roberts, 1963



Rosenfeld, 1969

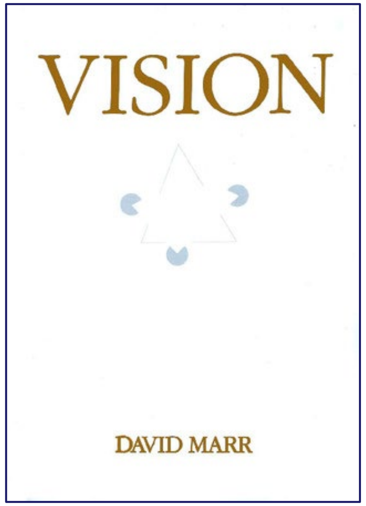


Duda & Hart, 1972

Origins of computer vision

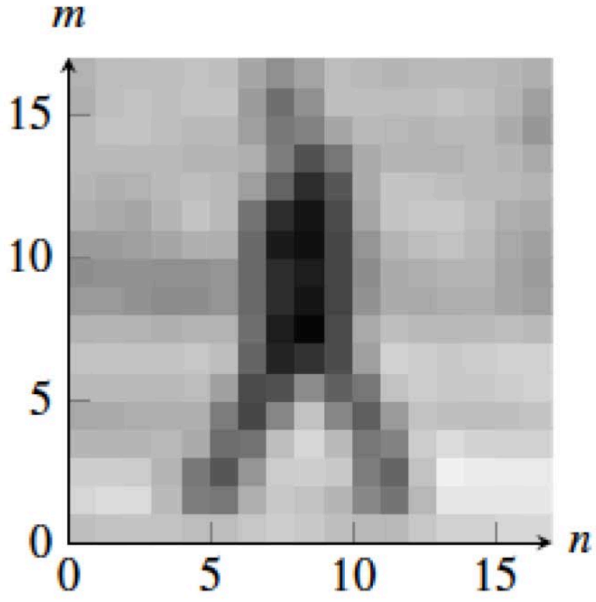
The first great revelation was that the problems are difficult. Of course, these days this fact is a commonplace. But in the 1960s almost no one realized that machine vision was difficult. The field had to go through the same experience as the machine translation field did in its fiascoes of the 1950s before it was at last realized that here were some problems that had to be taken seriously. The reason for this misperception is that we humans are ourselves so good at vision. The notion of a feature detector was well established by Barlow and by Hubel and Wiesel, and the idea that extracting edges and lines from images might be at all difficult simply did not occur to those who had not tried to do it. It turned out to be an elusive problem: Edges that are of critical importance from a three-dimensional point of view often cannot be found at all by looking at the intensity changes in an image. Any kind of textured image gives a multitude of noisy edge segments; variations in reflectance and illumination cause no end of trouble; and even if an edge has a clear existence at one point, it is as likely as not to fade out quite soon, appearing only in patches along its length in the image. The common and almost despairing feeling of the early investigators like B.K.P. Horn and T.O. Binford was that practically anything could happen in an image and furthermore that practically everything did.

David Marr (1945-1980)



Goal of this class: To get to know the pixels

What we see



What the machine gets

$$\mathbf{I} = \begin{bmatrix}
160 & 175 & 171 & 168 & 168 & 172 & 164 & 158 & 167 & 173 & 167 & 163 & 162 & 164 & 160 & 159 & 163 & 162 \\
149 & 164 & 172 & 175 & 178 & 179 & 176 & 118 & 97 & 168 & 175 & 171 & 169 & 175 & 176 & 177 & 165 & 152 \\
161 & 166 & 182 & 171 & 170 & 177 & 175 & 116 & 109 & 169 & 177 & 173 & 168 & 175 & 175 & 159 & 153 & 123 \\
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135 & 132 & 131 & 125 & 115 & 129 & 132 & 74 & 54 & 41 & 104 & 156 & 152 & 156 & 164 & 156 & 141 & 144 \\
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184 & 187 & 192 & 182 & 124 & 32 & 109 & 168 & 171 & 167 & 163 & 51 & 105 & 203 & 209 & 203 & 210 & 205 \\
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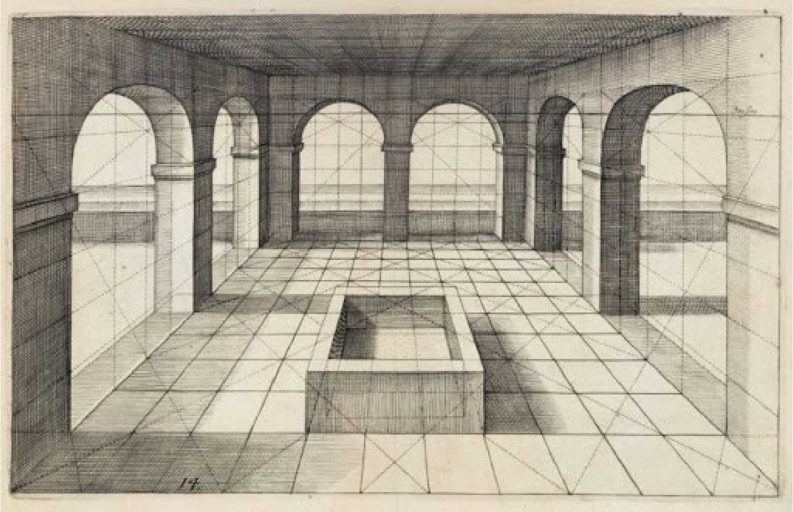
153 & 149 & 153 & 155 & 173 & 182 & 179 & 177 & 182 & 177 & 182 & 185 & 179 & 177 & 167 & 176 & 182 & 180
\end{bmatrix}$$

The Computer Vision Class in a Snapshot

Topics Studied in This Compute Vision Class

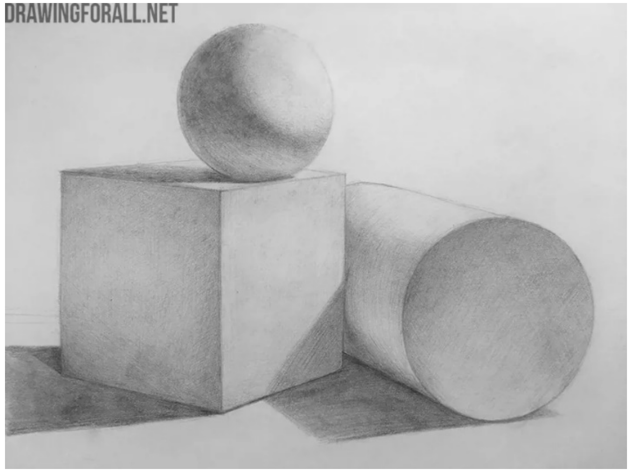
I. Image Formation
II. Image processing and early vision
III. Fitting and alignment
IV. 3D vision
V. Deep Learning for Computer Vision
VI. Applications

I. Image formation

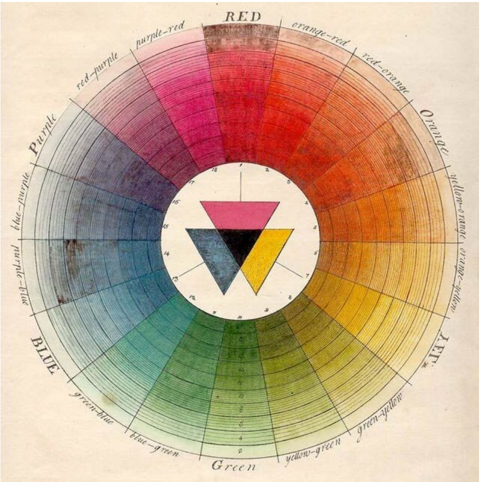


Perspective projection

Camera optics

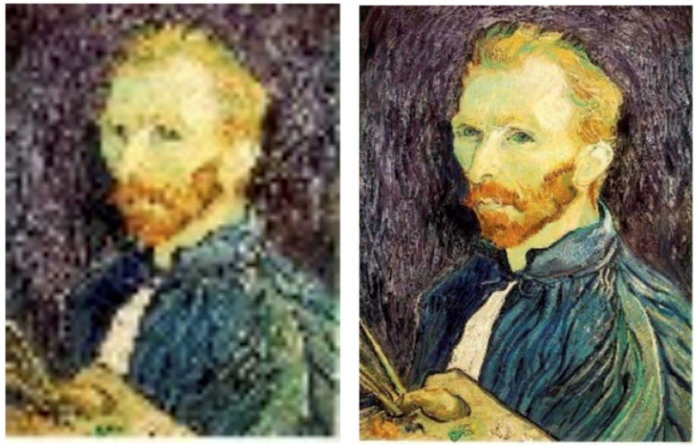


Light and shading

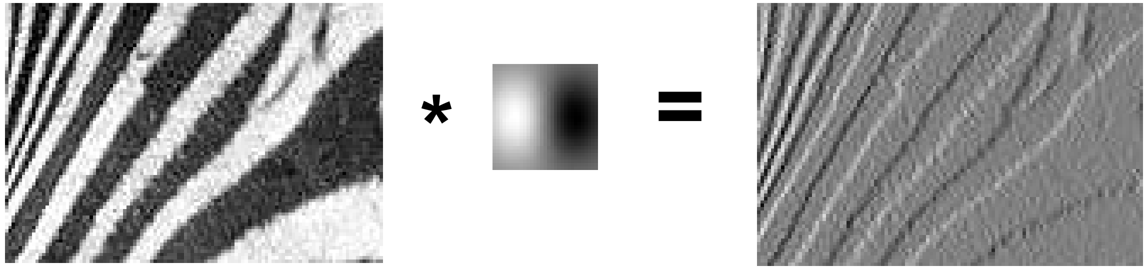


Color

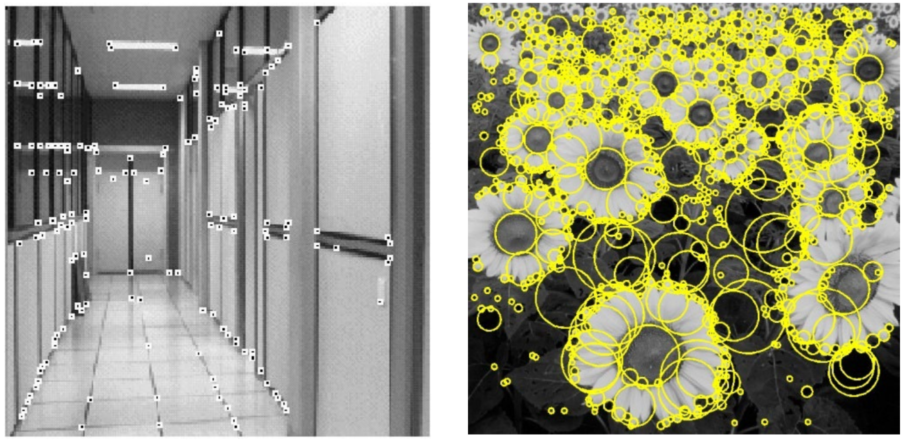
II. Image processing and early vision



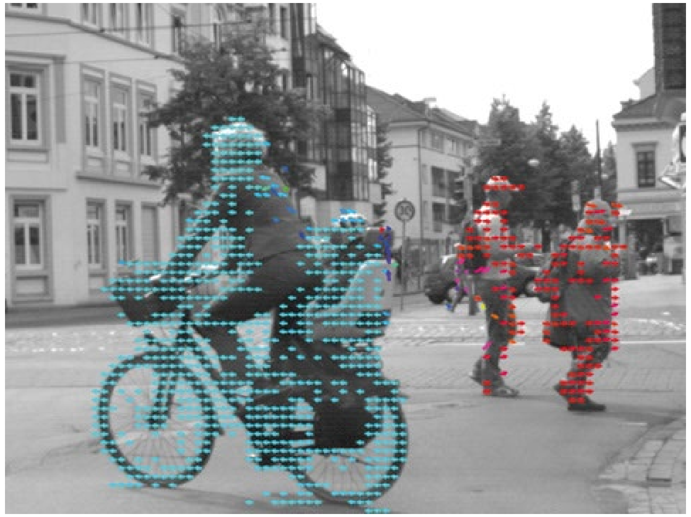
Sampling, interpolation, Fourier analysis



Linear filtering
Edge detection

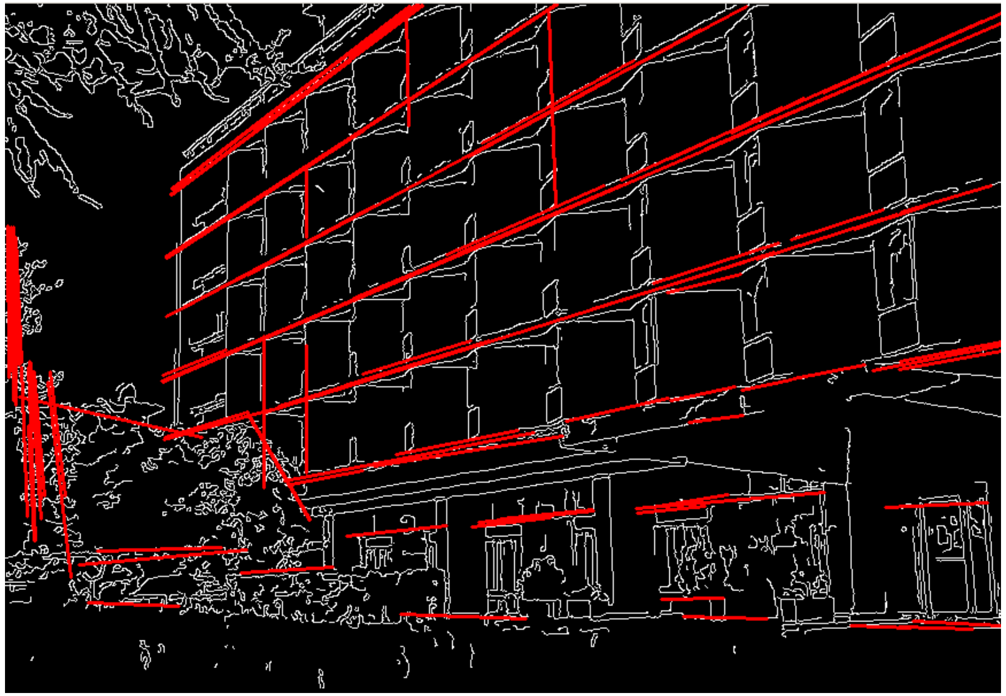


Feature extraction



Optical flow

III. Fitting and alignment

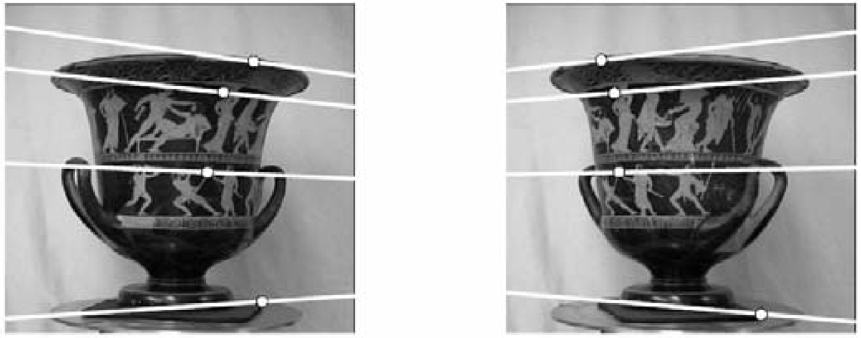


Fitting: Least squares, voting methods

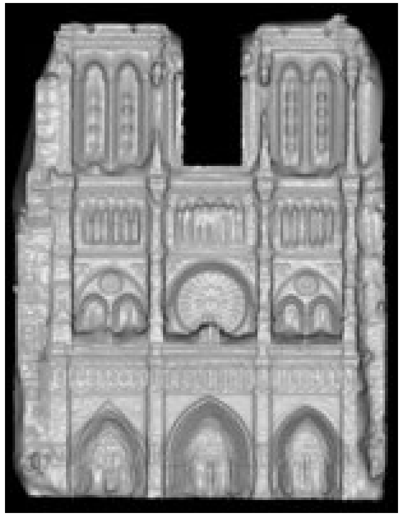
Alignment, image stitching

IV. 3D vision

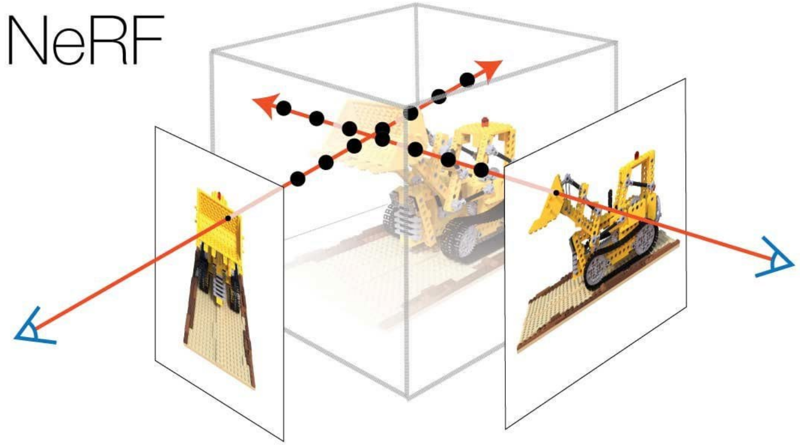
Camera calibration



Two-view geometry, structure from motion

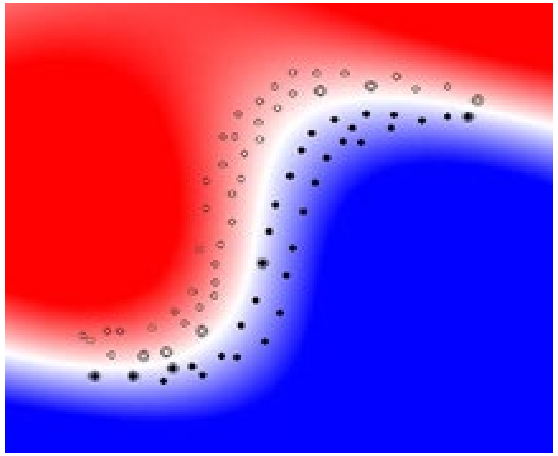


Stereo

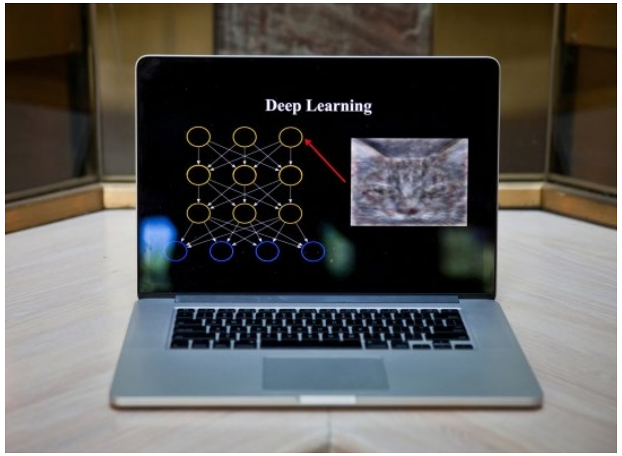


Light fields and dense 3D reconstruction

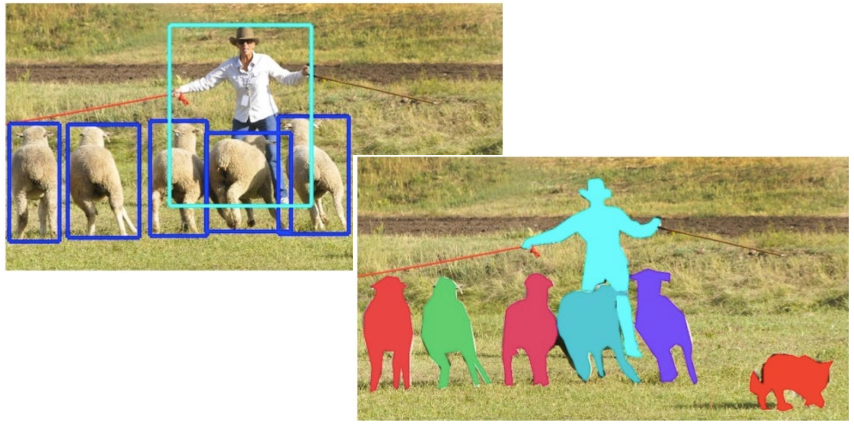
V. Deep Learning for Computer Vision



Intro to classification
and neural networks



Deep learning architectures for images



Object detection and segmentation

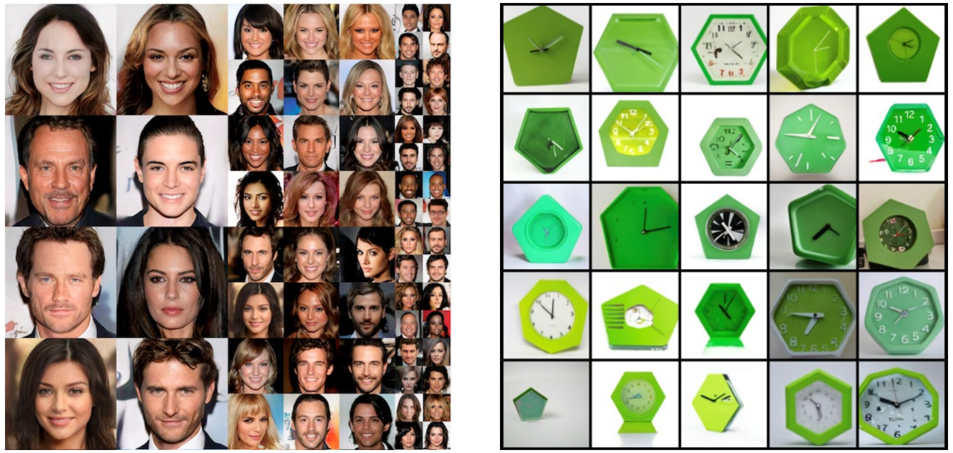


Image generation

Thank You!